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# Learning Sample-Aware Threshold for Semi-Supervised Learning

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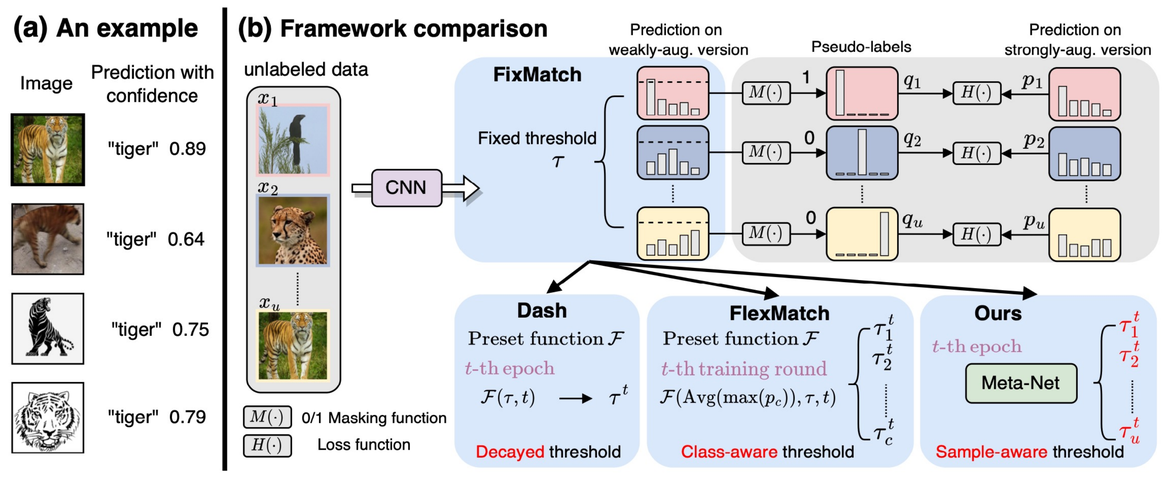
## Contributions

- A simple yet effective training framework called Meta-Threshold (Meta-T), which
  - does not leverage prior knowledge to preset adjust function for thresholds
  - contains one hyperparameter, thus does not require complex cross-validation.
- Theoretically provide the convergence of Meta-T which enjoys a rate of $\mathcal{O}(1/\epsilon^2)$.
- Meta-T be applied to solve both the conventional and imbalanced SSL tasks.

## Motivation and Framework




**(a) Motivation:** deep models have different learning capabilities for different examples in class *tiger*. Intuitively, setting instance-level thresholds is more logical and beneficial to generate more accurate pseudo-labels for unlabeled instances, further facilitating deep model's learning.

**(a) Review of the pseudo-labeling training framework:** Meta-T designs a meta-net which dynamically generates a refined confidence threshold for unlabeled example.

## Method

### Confidence Thresholds in Semi-Supervised Learning

Given an unlabeled data $\mathrm{x_m}$ , the training objective is

$$\ell_{\mathrm{x_m}} = 1(\max(f(\mathrm{A}^{\omega}(\mathrm{x_m}); \boldsymbol{w})) > \tau) \cdot H(\hat{y}_m, f(\mathrm{A}^{s}(\mathrm{x_m}); \boldsymbol{w}))$$

$\mathrm{A}^{\omega}$ weak augmentation
$\mathrm{A}^{s}$ strong augmentation
$H$ loss function
$\tau$ confidence threshold

### Meta-Threshold

The training objective in Meta-T is

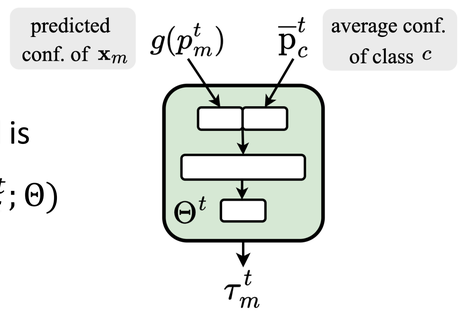
$$\ell_{\mathrm{x_m}} = 1(\max(f(\mathrm{A}^{\omega}(\mathrm{x_m}); \boldsymbol{w})) > \tau_m) \cdot H(\hat{y}_m, f(\mathrm{A}^{s}(\mathrm{x_m}); \boldsymbol{w}))$$

Sample-level threshold is produced by a meta-net $\tau_m = V_m(\boldsymbol{w}, \Theta))$

- **Structure of the meta-net**

Given $\mathrm{x_m}$, at epoch $t$, the generated threshold is

$$\tau_m^t = V(g(f(\mathrm{x_m}; \boldsymbol{w})), \bar{\mathrm{p}}_c^t; \Theta)$$




- **Bi-level optimization**

The optimal parameters of two networks can be obtained by minimizing the loss:

$$\mathbf{w}^*(\Theta) = \arg\min_{\mathbf{w}} L_u = \frac{1}{M} \sum\nolimits_{\mathbf{x}_m \in D^u} \ell_{\mathbf{x}_m}(\mathbf{w}, \Theta)$$

$$\Theta^* = \arg\min_{\Theta} L_{\mathrm{meta}}(\mathbf{w}^*(\Theta)) = \frac{1}{N} \sum\nolimits_{i=1}^{N} H_i(\mathbf{w}^*(\Theta))$$

Meta-optimization problem contains three steps:

(1) Formulating learning manner of classifier network

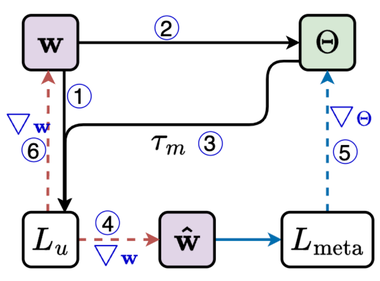
$$\hat{\mathbf{w}}^{(t)}(\Theta) = \mathbf{w}^{(t)} - \alpha \frac{1}{n\mu} \sum\nolimits_{i=1}^{n\mu} \nabla_{\mathbf{w}} \ell_{\mathbf{x}_i}(\mathbf{w}^{(t)}, \Theta^{(t)})$$

(2) Updating parameters Θ of TGN

$$\Theta^{(t+1)} = \Theta^{(t)} - \psi \frac{1}{n} \sum\nolimits_{i=1}^{n} \nabla_{\Theta} H_i(\hat{\mathbf{w}}^{(t)}(\Theta))$$

(3) Updating parameters $\boldsymbol{w}$ of classifier network

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \alpha \frac{1}{n\mu} \sum\nolimits_{i=1}^{n\mu} \nabla_{\mathbf{w}} \ell_{\mathbf{x}_i}(\mathbf{w}^{(t)}, \Theta^{(t+1)})$$




**Flowchart of Meta-T**

- **Learning algorithm**

**Algorithm 1** Learning algorithm of **Meta-T**.

**Require:** Unlabeled/labeled data $D^u/D^l$, batch size $n$, a coefficient $\mu$, max iterations $T$.
**Ensure:** Classifier network parameter $\mathbf{w}^{(T)}$.
1: Initialize $\mathbf{w}^{(0)}$ for classifier network and $\Theta^{(0)}$ for TGN.
2: **for** $t = 0$ **to** $T-1$ **do**
3: Random sample $\{(\mathbf{x}_1^l, \mathbf{y}_1^l), ..., (\mathbf{x}_n^l, \mathbf{y}_n^l)\}$ from $D^l$ and $\{\mathbf{x}_1, ..., \mathbf{x}_{(\mu \times n)}\}$ from $D^u$.
4: Calculate $\hat{\mathbf{w}}^{(t)}(\Theta)$. ▷ Eq. (6)
5: Update $\Theta^{(t+1)}$. ▷ Eq. (7)
6: Update $\mathbf{w}^{(t+1)}$. ▷ Eq. (8)
7: **end for**

## Experiments

### SOTA performance on eight test benchmarks (typical SSL)

| Methods | CIFAR-10 (Wide ResNet-28-2) 40 labels | 250 labels | 4000 labels | CIFAR-100 (Wide ResNet-28-8) 400 labels | 2500 labels | 10000 labels |
|---|---|---|---|---|---|---|
| Π-Model | - | 54.26±3.97 | 14.01±0.38 | - | 57.25±0.48 | 37.88±0.11 |
| VAT | 74.66±2.12 | 41.03±1.79 | 10.51±0.12 | 85.20±1.40 | 46.84±0.79 | 32.14±0.19 |
| MixMatch | 47.54±11.50 | 11.05±0.86 | 6.42±0.10 | 67.61±1.32 | 39.94±0.37 | 28.31±0.33 |
| UDA | 29.05±5.93 | 8.82±1.08 | 4.88±0.18 | 59.28±0.88 | 33.13±0.22 | 24.50±0.25 |
| CoMatch | 6.91±1.39 | 4.91±0.33 | - | - | - | - |
| SimMatch | 5.60±1.37 | 4.84±0.39 | **3.96±0.01** | 37.81±2.21 | **25.07±0.32** | **20.58±0.11** |
| Pseudo-labeling | - | 49.78±0.43 | 16.09±0.28 | - | 57.38±0.46 | 36.21±0.19 |
| FixMatch | 11.39±3.37 | 5.07±0.65 | 4.26±0.05 | 48.85±1.75 | 28.29±0.11 | 22.60±0.12 |
| Dash | 9.16±4.31 | 4.78±0.12 | 4.13±0.06 | 44.83±1.36 | 27.18±0.21 | 21.97±0.14 |
| FlexMatch | 4.97±0.06 | 4.98±0.09 | 4.19±0.01 | 39.94±1.62 | 26.49±0.20 | 21.90±0.15 |
| **Meta-T** (ours) | **4.39±0.28** | **4.10±0.20** | 4.01±0.09 | **36.17±1.40** | 25.81±0.72 | 20.74±0.23 |

| Error rates (%) ↓ Methods | SVHN 40 labels | 250 labels | STL-10 1000 labels |
|---|---|---|---|
| Π-Model | - | 18.96±1.92 | 26.23±0.82 |
| VAT | 74.75±3.38 | 4.33±0.12 | 37.95±1.12 |
| MixMatch | 42.55±14.53 | 3.98±0.23 | 10.41±0.61 |
| UDA | 52.63±20.51 | 5.69±2.76 | 7.66±0.56 |
| ReMixMatch | 3.34±0.20 | 2.92±0.48 | 5.23±0.45 |
| PL | - | 20.21±1.09 | 27.99±0.83 |
| FixMatch | 3.14±1.60 | 2.64±0.64 | 5.17±0.63 |
| Dash | 3.03±1.59 | **2.17±0.10** | 3.96±0.25 |
| FlexMatch | 8.19±3.20 | - | 5.77±0.18 |
| **Meta-T** (ours) | **2.89±0.92** | 2.29±0.51 | **3.51±0.34** |

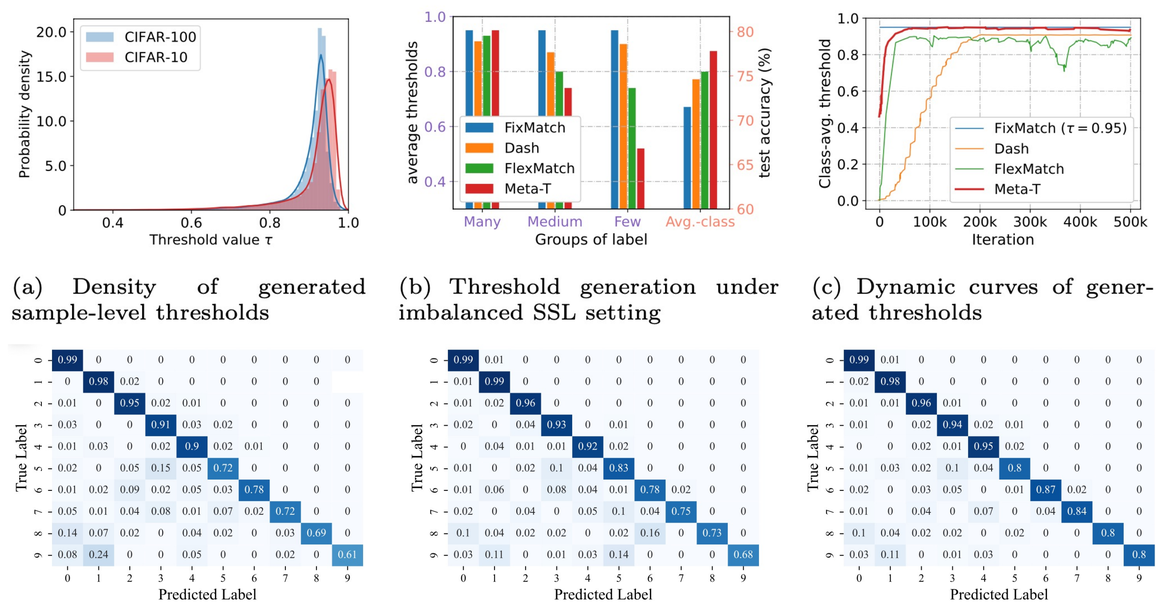
| Top-1 / Top-5 accuracy (%) ↑ | ImageNet 1% | 10% | 100% |
|---|---|---|---|
| Sup. baseline | 25.4 / 48.4 | 56.4 / 80.4 | |
| FixMatch | 53.4 / 74.4 | 70.8 / 89.0 | |
| CoMatch | 66.0 / 86.4 | 73.6 / 91.6 | 80.4 / 94.6 |
| SimMatch | 67.2 / 87.1 | 74.4 / 91.6 | |
| **Meta-T** (ours) | **67.7 / 87.9** | **75.0 / 91.7** | |

| Error rates (%) ↓ | IMDb | Amazon-5 | Yelp-5 |
|---|---|---|---|
| UAD | 18.33±0.61 | 50.29±4.6 | 47.49±6.83 |
| FixMatch | 7.59±0.28 | 42.70±0.53 | 39.56±0.70 |
| FlexMatch | 7.80±0.23 | 42.34±0.62 | 39.01±0.17 |
| SoftMatch | 7.48±0.12 | **42.14±0.92** | 39.31±0.45 |
| **Meta-T**(ours) | **7.20±0.20** | 42.60±0.41 | **38.44±0.37** |

### SOTA performance on imbalanced SSL task

| Methods | $N_1 = 1500, M_1 = 3000$ $\gamma = 50$ | $\gamma = 100$ | $\gamma = 150$ | $N_1 = 500, M_1 = 4000$ $\gamma = 50$ | $\gamma = 100$ | $\gamma = 150$ |
|---|---|---|---|---|---|---|
| Supervised | 65.23±0.05 | 58.94±0.13 | 55.63±0.38 | 51.31±0.34 | 45.82±0.41 | 40.90±0.39 |
| cRT | 67.82±0.14 | 63.43±0.45 | 59.56±0.44 | 56.28±1.45 | 48.11±0.79 | 45.02±1.08 |
| LDAM | 68.91±0.10 | 63.15±0.24 | 58.68±0.30 | 56.41±0.92 | 49.27±0.88 | 45.10±0.75 |
| MixMatch | 73.59±0.46 | 65.03±0.26 | 62.71±0.29 | 65.32±1.20 | 56.41±1.96 | 52.38±1.88 |
| ReMixMatch | 78.96±0.29 | 72.88±0.12 | 68.61±0.40 | 76.83±0.98 | 70.12±1.23 | 59.58±1.30 |
| DARP | 81.60±0.31 | 75.23±0.14 | 69.31±0.26 | 76.72±0.46 | 69.41±0.50 | 61.23±0.31 |
| CReST | 82.03±0.26 | 75.08±0.41 | 69.84±0.39 | 76.18±0.36 | 69.50±0.70 | 60.81±0.55 |
| Adsh | 83.38±0.06 | 76.52±0.35 | 71.49±0.30 | **79.27±0.38** | 70.97±0.46 | 62.04±0.51 |
| FixMatch | 79.10±0.14 | 71.50±0.31 | 68.47±0.15 | 77.34±0.96 | 68.45±0.94 | 60.10±0.82 |
| Dash | 81.93±0.10 | 74.62±0.26 | 72.29±0.42 | 77.90±0.39 | 70.41±0.27 | 62.11±0.32 |
| FlexMatch | 82.86±0.25 | 75.47±0.41 | 70.62±0.30 | 78.69±0.50 | 71.80±0.29 | 62.85±0.39 |
| **Meta-T** (ours) | **83.94±0.12** | **77.80±0.39** | **73.07±0.58** | 78.41±0.22 | **72.40±0.42** | **64.46±0.60** |

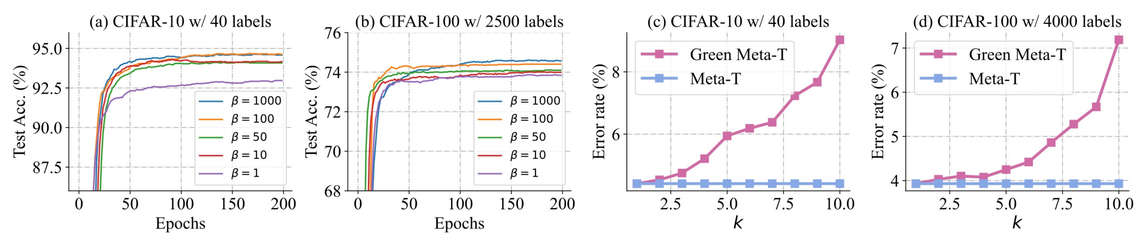
### Effectiveness analysis




(a) Density of generated sample-level thresholds (b) Threshold generation under imbalanced SSL setting (c) Dynamic curves of generated thresholds

(a) FixMatch (b) FlexMatch (c) Meta-T

### Sensitivity analysis




## Reference

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